PA: Product and Presentation

Designed to redefine the reading experience, Liberato combines technology, accessibility, and community to create a modern space for book lovers. The platform enables readers to easily discover, purchase, and review books. By giving customers a seamless shopping experience, Liberato empowers readers, authors, and publishers to connect through a shared passion for literature.

A9: Product

> The Lierato final product represents the implementation of the information system conceptualised in the earlier stages (A1 to A7), using PHP and The Laravel Framework to create dynamic web pages, AJAX for enhanced user experience, and PostgreSQL as the database solution. Designed to redefine the reading experience, Liberato combines technology, accessibility, and community to create a modern space for book lovers. The platform enables readers to easily discover, purchase, and review books.

1. Installation

> Link to the release with the final version of the source code in the group's git repository: https://gitlab.up.pt/lbaw/lbaw2526/lbaw25134

> Full Docker command to launch the image available in the group's GitLab Container Registry using the production database:
> `docker run -d --name lbaw25134 -p 8001:80 gitlab.up.pt:5050/lbaw/lbaw2526/lbaw25134`

2. Usage

2.1. Administration Credentials

> Administration URL: <URL>

Email	Password
admin1@liberato.com	password

2.2. User Credentials

Type	Email	Password
regular user	john@example.com	password
seller	carlos@example.com	password

3. Application Help

Help has been implemented across various features to provide users with clear feedback on their actions:

Authentication & Account Management

- Registration and login validation with clear error messages
- Password recovery with confirmation and error feedback
- Session management and logout confirmations
- Access restrictions for blocked accounts

Shopping & Orders

- Cart operations with instant feedback (add/remove items, quantity updates)
- Stock availability warnings and validation
- Checkout validation (payment method selection, stock checks)
- Order confirmation with complete details
- Purchase history and order status notifications

Product Interaction

- Review submission with validation (rating 0-5 stars, purchase requirement)
- Wishlist toggle feedback
- Search and filter results with "no results" guidance
- Category and price filter indicators

Notifications

- Real-time notification badge updates
- Mark as read/unread confirmations
- Order status, price changes, and availability alerts
- Empty state guidance with suggested actions

Profile Management

- Profile update confirmations
- Image upload validation (file type, max 2MB)
- Form field validation (email, username, password)
- Duplicate prevention messages

Admin & Seller Features

- CRUD operation success/error messages
- Role-based access restrictions
- Stock and inventory management feedback
- Relationship constraint warnings (e.g., cannot delete category with books)
- User role assignment confirmations

Visual example:

4. Input Validation

Input validation is implemented on both client-side and server-side to ensure data integrity and security:

Client-Side Validation: Uses HTML5 attributes and JavaScript for instant user feedback:

- `type="email"`, `type="tel"`, `type="number"` for format enforcement
- `required` attribute for mandatory fields
- `min`, `max`, `maxlength` for value constraints
- Real-time password confirmation matching
- File type and size validation (images only, max 2MB)

Server-Side Validation: Laravel validation rules ensure security and data integrity:

Authentication & User Management

```
// Registration
$request->validate([
    'username' => 'required|string|max:250|unique:users',
    'email' => 'required|email|unique:users',
    'password' => 'required|string|min:8|confirmed'
```

```
]);

// Profile update with uniqueness check excluding current user
$request->validate([
    'username' => 'required|unique:users,username,' . auth()->id(),
    'profile_img' => 'nullable|image|mimes:jpeg,png,jpg,gif|max:2048'
]);
```

E-commerce Operations

```
// Payment method validation
if ($paymentMethod === 'Card') {
    $rules['card_number'] = 'required|digits:16';
    $rules['card_cvc'] = 'required|digits_between:3,4';
} elseif ($paymentMethod === 'MBWay') {
    $rules['mbway_phone'] = 'required|regex:/^9[0-9]{8}$/';
}

// Cart and stock validation
$request->validate([
    'book_id' => 'required|exists:book,book_id',
    'quantity' => 'required|integer|min:1'
]);
// Additional stock availability check
```

Content Management (Admin/Seller)

```
// Book management
$request->validate([
    'name' => 'required|string|max:255',
    'price' => 'required|numeric|min:0',
    'available_stock' => 'required|integer|min:1',
    'category_name' => 'required|exists:category,category_name'
]);

// Category with uniqueness and relationship checks
$request->validate([
    'category_name' => 'required|unique:category,category_name'
]);
// Prevent deletion if books exist in category
```

Additional Validations

- Product reviews: Rating 0-5, user must have purchased book
- Search queries: Max length 255, special character sanitization
- Price ranges: Min/max numeric validation, logical consistency
- User roles: Boolean validation for is_seller, is_blocked fields

All validation errors display contextually near form fields with clear, actionable messages. Global success/error messages appear at the page top, while field-specific errors show inline.

5. Check Accessibility and Usability

Accessibility and usability tests were performed using the checklists provided in these links:

- https://ux.sapo.pt/checklists/acessibilidade/
- https://ux.sapo.pt/checklists/usabilidade/

> Accessibility

> Usability

6. HTML & CSS Validation

HTML and CSS validation results for the central page of the product:

> HTML:

Home catalogue page: validation pdf file here
Admin dashboard page: validation pdf file here
Checkout page: validation pdf file here

> CSS: app.css_validation here

7. Revisions to the Project

Since the requirements specification stage, some changes were made to the project:

ER - Fixed errors in user stories, reorganised by priority, removed system actor (unnecessary), and adjusted user profiles from A1.
EBD - SQL script now cleans the database (DROP tables), UML revised with correct generalizations/associations (specialized notifications, seller-book 1:N, payment-order 1:1 with PK in payment, admin-category, reviews M:N with proper relation, RESOLVES mapped, wishlist as association class, username key in R1), Status domain created, auto PKs, creation script corrected and production DB aligned.
EAP - Added AJAX requests.

8. Implementation Details

8.1. Libraries Used

Libraries and frameworks used in the product:

Laravel 12: PHP framework for MVC architecture, routing, authentication, database ORM (Eloquent), session management, and password reset logic
PostgreSQL 16: Relational database with full-text search
Mailtrap: SMTP service for testing password recovery emails in development
Docker & Docker Compose: Local PostgreSQL + pgAdmin containerization
Nginx: Web server and reverse proxy for PHP-FPM in production deployment
Laravel Pail: Real-time log viewer integrated in development script
Concurrently: NPM tool for running multiple dev processes simultaneously

8.2 User Stories

> This subsection includes all user stories, sorted by order of implementation, with the owner of the user story with their name in bold.

US Identifier	Name	Module	Priority	Team Members	State
US01	Registration	M01: Authentication	High	**João Santos**	100%
US02	Login	M01: Authentication	High	**João Santos**	100%
US15	Placeholders in Form Inputs	Module	Innovation	**João Santos**, Iara Brás, Dinis Pinto, Ricardo Fernandes	100%
US28	Contextual Error Messages	Module	Medium	**João Santos**, Iara Brás, Dinis Pinto, Ricardo Fernandes	100%
US29	Contextual Help	Module	Medium	**João Santos**, Iara Brás, Dinis Pinto, Ricardo Fernandes	80%
US32	Log Out	M01: Authentication	Medium	**João Santos**	100%
US17	Main Features	M12: Home & Platform Info	Innovation	**João Santos**, Iara Brás, Dinis Pinto, Ricardo Fernandes	100%
US03	View & Search Products	M03: Catalog	High	**João Santos**	100%
US13	Browse Product Categories	M03: Catalog	Medium	**João Santos**	100%

US Identifier	Name	Module	Priority	Team Members	State
US11	Search Filters	M07: Search & Discovery	Medium	**João Santos**	100%
US04	Exact Match Search	M07: Search & Discovery	High	**João Santos**, Dinis Pinto	100%
US05	Full-text Search	M07: Search & Discovery	High	**João Santos**, Dinis Pinto	100%
US08	View Product Details	Module	High	**João Santos**, Iara Brás	100%
US33	View Product Details & Reviews	Module	Medium	**João Santos**	100%
US19	View Profile	M02: Profile	High	**João Santos**	100%
US20	Edit Profile	M02: Profile	High	**João Santos**	100%
US26	Upload/Update Profile Picture	M02: Profile	Medium	**Dinis Pinto**	100%
US06	Add item to Cart	M04: Shopping Cart	High	**Iara Brás**	100%
US07	Manage Cart	M04: Shopping Cart	High	**Iara Brás**	100%
US21	Checkout	M05: Checkout & Orders	High	**Iara Brás**	100%
US34	Manage Purchases	M05: Checkout & Orders	Medium	**Ricardo Fernandes**	100%
US09	View Product Reviews	M08: Reviews & Ratings	Medium	**João Santos**	100%
US10	Search over Multiple Attributes	M07: Search & Discovery	Medium	**João Santos**	100%
US14	Faceted Search	M07: Search & Discovery	Innovation	**João Santos**	100%
US12	Ordering of Results	M07: Search & Discovery	Medium	**João Santos**	100%
US47	Support Admin Accounts	M10: Administration	High	**João Santos**, Iara Brás, Dinis Pinto, Ricardo Fernandes	100%
US48	Administer User Accounts	M10: Administration	High	**João Santos**	100%
US49	Add Product	M10: Administration	High	**João Santos**	100%
US50	Manage Products Information	M10: Administration	High	**Ricardo Fernandes**	100%
US54	Manage Products Stock	M10: Administration	Medium	**Ricardo Fernandes**	100%
US52	Block/Unblock User Accounts	M10: Administration	Medium	**João Santos**	100%
US35	Manage Wishlist	Module	Medium	**João Santos**	100%
US22	Review Purchased Product	M08: Reviews & Ratings	High	**João Santos**	100%

US Identifier	Name	Module	Priority	Team Members	State
US36	Write Reviews	M08: Reviews & Ratings	Medium	**João Santos**	100%
US37	Edit Review	M08: Reviews & Ratings	Medium	**João Santos**	100%
US38	Remove Review	M08: Reviews & Ratings	Medium	**João Santos**	100%
US16	About Us	M12: Home & Platform Info	Innovation	**Dinis Pinto**	100%
US18	Contacts	M12: Home & Platform Info	Innovation	**Dinis Pinto**	100%
US45	Manage their stock	M11: Seller Tools	High	**Ricardo Fernandes**	100%
US44	Appeal for Unblock	M02: Profile	Innovation	**Ricardo Fernandes**	100%
US46	Manage their sales stats	M11: Seller Tools	Medium	**Ricardo Fernandes**	100%
US27	View Personal Notifications	M09: Notifications	Medium	**Iara Brás**	100%
US30	View Notification List	M09: Notifications	Medium	**Iara Brás**	100%
US31	Mark Notifications as Read	M09: Notifications	Medium	**Iara Brás**	100%
US23	Payment Approved	M09: Notifications	High	**Iara Brás**	100%
US43	Product on Cart Price Change	M09: Notifications	Medium	**Iara Brás**	100%
US42	Product in Wishlist Available	M09: Notifications	Medium	**Iara Brás**	100%
US24	Recover Password	M02: Profile	Medium	**Dinis Pinto**	100%
US55	Manage Product Categories	M10: Administration	Medium	**Iara Brás**	100%
US56	View Users' Purchase History	M10: Administration	Medium	**Iara Brás**	100%
US53	Delete User Account (Admin)	M10: Administration	Medium	**Dinis Pinto**	100%
US25	Delete Account	M02: Profile	Medium	**Dinis Pinto**	100%

Not implemented:

US Identifier	Name	Priority	Team Member	Description
US39	Track Order	Medium	Dinis	As an authenticated user, I want to track my order status, so that I can know when my purchase will arrive.
US40	Cancel Order	Medium	João Santos	As an authenticated user, I want to cancel my order, so that I can stop the purchase if I change my mind or find an issue.

US Identifier	Name	Priority	Team Member	Description
US41	Change in Order Processing Stage	Medium	Dinis	As an authenticated user, I want to be notified when the processing stage of my order changes, so that I can track its progress.
US51	Manage Order Status	High	João	As an admin, I want to update and manage the status of orders, so that I can oversee order processing and fulfilment.

A10: Presentation

> This is the product presentation artifact.

1. Product presentation

> Brief presentation of the product and its main features:

Liberato is an online store designed for book lovers. Its main goal is to make it easy for users to find, buy, and review books. The platform has all the features you'd expect from a retail website: you can sign up and log in, search for products using filters, add items to a shopping cart, and complete the checkout process. Registered users can also manage their personal profile, track their orders, and save products they like to a wishlist.

The project also includes features for different types of users who manage the site. Sellers can add products, manage their stock levels, and view their sales data. Administrators have high-level control to manage all users, fix or update product details and stock, and oversee the status of all customer orders. Essentially, the system is a comprehensive bookstore with intuitive shopping tools and detailed controls to ensure the site operates smoothly and securely.

2. Video presentation

> Link to the lbaw25134.mp4 file:

> Screenshot of the video:

Revision history

Changes made to the first submission:

1. Item 1
2. ..

GROUP25134, 22/12/2025

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Comments